Whole−tree ASR − AIC−weighted average (gr6 0.445; gr7 0.555)

Haroldius sp2 STL10273
Haroldius sp1 STL10266
Dicranocara deschodti STL10269
Byrrhidium namaquensis STL10268
Endroedyolus paradoxus STL10049
Odontoloma sp1 STL10288
Odontoloma planatum STL10048
Paraphytus cf hindu STL10093
Frankenbergerius armatus STL10322
Coptorhina klugii STL10039
Sarrophorus costatus ST006
Ochichanton sp1 STL25
Grebennikovius lupanganus sp n1 CNCCOLVG00004076
Epactoides sp3 STL10162
Epactoides mangabeensis STL10154
Epactoides olsoufieffi STL10141
Epactoides aff Semiaeneus STL10158
Epactoides semiaeneus STL10136
Epactoides aff Lissus STL10134
Epactoides sp2 STL10157
Epactoides hanski STL29
Epactoides major STL47
Epactoides aff major STL10163
Epactoides femoralis STL10145
Epactoides jounii STL33
Epactoides sp STL5017
Bohepilissus sp1 STL10279
Epirinus cf.silvestris STL10018
Epirinus flagellatus STL10118
Epirinus sp. 1 STL43
Nesosisyphus pygmaeus STL3
Nesosisyphus vicinus STL1
Nesosisyphus rotundatus STL52
Nesosisyphus regnardi STL2
Sisyphini sp morpho1 STL18
Sisyphus muricatus STL5
Sisyphus manni STL15
Sisyphus gazanu s STL9
Sisyphus inconspicuus STL16
Sisyphus sordidus STL6
Sisyphus australis STL13
Sisyphus schaefferi STL44
Sisyphini sp morpho8 STL40
Sisyphini sp morpho6 STL23
Sisyphini sp morpho 10 STL42
Sisyphus umbraphilus STL10
Sisyphus (Neosisyphus) mirabilis STL11
Sisyphus (Neosisyphus) calcaratus STL7
Sisyphus (Neosisyphus) barbarossa STL8
Sisyphus (Neosisyphus) rubrus STL12
Sisyphus (Neosisyphus) spinipes STL17
Sisyphus (Neosisyphus) infuscatus STL14
Panelus rhodesiensis STL10037
Panelus sp3 STL51
Janssensantus mabuensis n sp STL10102
Janssensantus cf inexpectatus STL10103
Tanzanolus sp1 STL10223
Nesovinsionia vinsoni STL4
Paragymnopleurus sp STL10293
Gymnopleurus aenescens STL10097
Paragymnopleurus sp STL10092
Garreta unicolor STL10077
Gyronotus pumilus STL10003
Chalconotus convexus STL10294
Circellium bacchus STL10270
Kheper nigroaeneus STL10036
Pachysoma hippocrates STL10127
Scarabaeus n sp1 STL10210
Pachylomera femoralis STL10128
Drepanopodus proximus STL10001
Sceliages hippias STL10030
Scarabaeus westwoodi STL10034
Scarabaeolus intricatus STL10044
Nanos aff Pseudominutus STL10156
Nanos sp1 STL10147
Nanos aff Bicoloratus STL10137
Nanos bicoloratus STL10155
Cambefortantus pauliani STL32
Nanos aff humeralis STL10190
Nanos sp5 STL10202
Nanos sp4 STL10215
Nanos sp2 STL10153
Arachnodes aff metallicus STL10196
Apotolamprus sp. morpho 1 STL31
Apotolamprus perinetensis STL10161
Apotolamprus sp1 STL10146
Apotolamprus aff metallicus STL10160
Apotolamprus aff sparsepunctatus STL10198
Apotolamprus aff latipennis STL10197
Apotolamprus quadrinotatus STL5006
Apotolamprus aff ambohitsitondronensi STL10140
Apotolamprus quadrimaculatus STL10139
Apotolamprus vadoni STL10135
Nanos vadoni ST011
Nanos aff rubrosignatus STL10142
Nanos sp6 STL5002
Nanos sp STL39
Nanos sp9 STL5004
Nanos sp11 STL10207
Nanos sp7 STL5005
Nanos sp10 STL10206
Nanos sp8 STL10204
Nanos aff viridissimus STL10191
Apotolamprus sp STL30
Nanos aff ST009
Nanos dubitatus STL5001
Nanos aff mirjae STL10189
Nanos morpho 2 STL35
Synapsis sp STL10129
Heliocopris sp STL10292
Heliocopris sp STL10271
Temnoplectron lewisensis Temlew UCE
Diorygopyx dudplodentatus Diodud UCE
Amphistomus primonactus Amppri UCE
Thyregis relictus Thyrel UCE
Coptodactyla brooksi CopbroUCE
Onycothechus tricornigeris STL10225
Pedaria sp2 STL10019
Pedaria sp STL10323
Pedaria sp1 STL10016
Microcopris sp1 STL10023
Copris sl diversus STL10245
Litocopris muticus STL10027
Copris urus STL10024
Copris sp2 STL10025
Copris nevinsoni STL10021
Copris gilleti STL10028
Copris fidius ST005
Copris (Paracopris) sp. STL10022
Apterepilissus ovalis STL10325
Arachnodes oberthuri STL10144
Arachnodes sp6 STL10200
Arachnodes sp morpho 2 STL37
Arachnodes sp3 STL10218
Arachnodes sp1 STL5012
Arachnodes sp5 STL10217
Arachnodes aff robinsoni STL10194
Arachnodes sp2 STL5013
Arachnodes saprinoides STL10151
Arachnodes aff saprinoides STL10195
Arachnodes sp STL10216
Arachnodes sp. morpho 1 STL45
Arachnodes pusillus STL5015
Arachnodes sp4 STL5016
Arachnodes sp7 cf insularis STL10201
Epilissus aff semilissus STL10181
Epilissus splendidus cf delphinensis STL10185
Epilissus splendidus splendidus ST008
Epilissus sp morpho 2 STL48
Epilissus sp1 STL10186
Epilissus aff antoetrae STL10180
Epilissus apotolamproides STL5008
Epilissus aff planatus STL10187
Epilissus alluaudi STL5009
Epilissus fuscovirens STL5010
Epilissus morio STL5007
Epilissus aff morio group STL10184
Epilissus sp morpho 1 STL26
Epilissus cuprarius STL5011
Metacatharsius sp1 STL10007
Metacatharsius opacus STL10017
Catharsius cf molossus STL10120
Copridaspidus peregrinus STL10258
Catharsius sp. STL10033
Dwesasilvasedis medinae STL10265
Xinidium dewitzi STL10008
Xinidium dentilabris STL10004
Macroderes sp1 STL10328
Macroderes foveatus STL10040
Cheironitis scabrosus STL10096
Onitis uncinatus STL10031
Heteronitis ragazzii STL10032
Onitis caffer STL10042
Megalonitis bohemani STL10318
Anonychonitis freyi STL10095
Onitis alexis STL10041
Pycnopanelus krikkeni STL10076
Hammondantus psammophilus STL10321
Kurtops signatus STL10043
Hamonthophagus depressus STL10013
Onthophagus probus STL10015
Digitonthophagus gazella STL10213
Phalops flavocinctus STL10045
Amietina larrochei STL10208
Onthophagus Palaeonthophagus gibbulus STL10133
Onthophagus sl sp1 cf gr STL10315
Caccobius binodulus STL10300
Onthophagus sl lamelliger STL10131
Onthophagus Furconthophagus variegatus STL10083
Onthophagus sl vinctus STL10080
Onthophagus sl borneensis STL10306
Onthophagus Pseudophanaeomorphus quasijohkii aff STL10299
Onthophagus Euonthophagus flavimargo STL10109
Onthophagus Bicornonthophagus interstitialis STL10081
Cleptocaccobius viridicollis STL10099
Caccobius schereberi ST001
Onthophagus sl leucopygus STL10100
Onthophagus sl cf quadrimaculatus STL10046
Onthophagus cf.bicavifrons STL10006
Eusaproecius sp1 STL10256
Onthophagus sl giraffa STL10130
Onthophagus sl pilosus STL10101
Onthophagus sl baeri STL10084
Onthophagus taurus ST002
Onthophagus sstr obscurior STL10305
Onthophagus Gibbonthophagus rectecornutus STL10114
Onthophagus sl vulpes STL10307
Onthophagus Onthophagiellus cf sp1 STL10304
Onthophagus Micronthophagus sp1 STL10115
Onthophagus sl batillifer STL10301
Onthophagus Hikidaeus waterstradti STL10308
Onthophagus Paraphanaeomorphus aff phanaeiformis STL10113
Onthophagus pipitzi STL5018
Proagoderus sapphirinus STL10098
Proagoderus cf lanista STL10079
Onthophagus Parascatonomus tricornis STL10111
Parascatonomus anceyi STL10091
Onthophagus cf muelleri STL10298
Onthophagus Serrophorus seniculus STL10110
Onthophagus Sunenaga streltsovi cf STL10303
Onthophagus sl laevis STL10302
Macronthophagus diabolicus STL10291
Euoniticellus fulvus STL10132
Oniticellus cinctus ST004
Liatongus sp STL10290
Oniticellus planatus STL10119
Scaptocnemis segregis STL10251
Cyptochirus sp1 STL10112
Helictopleurus vadoni STL10244
Helictopleurus neoamplicollis STL5024
Helictopleurus viettei STL10241
Helictopleurus peyrierasi STL5021
Helictopleurus sicardi ST013
Helictopleurus carbonarius STL10249
Helictopleurus cambeforti STL10250
Helictopleurus semivirens STL10242
Helictopleurus cf steineri n sp STL10211
Helictopleurus steineri STL10264
Helictopleurus dorbignyi STL10240
Helictopleurus sp group semivirens STL10214
Helictopleurus nicollei STL10319
Helictopleurus marsyas STL5026
Helictopleurus coruscus STL10209
Helictopleurus aurocupreus STL5025
Helictopleurus sinuatocornis STL10330
Helictopleurus giganteus STL10177
Helictopleurus cribricollis STL10243
Helictopleurus undatus STL10331
Helictopleurus rudicollis STL10173
Helictopleurus perrieri STL10329
Helictopleurus quadripunctatus STL5027
Helictopleurus unifasciatus STL5030
Helictopleurus splendidicollis STL5020
Helictopleurus politicollis STL5029
Helictopleurus viridans pseudoviridans STL10236
Helictopleurus viridans viridans STL10235
Helictopleurus hanskii STL10247
Helictopleurus fulgens purpuricollis STL10239
Helictopleurus nigricans STL10238
Helictopleurus fulgens STL10168
Helictopleurus fissicollis STL28
Helictopleurus fissicollis australis STL10246
Helictopleurus fissicollis morettoi STL10237
Helictopleurus neuter STL10327
Helictopleurus fasciolatus obscurus STL10334
Helictopleurus fasciolatus pseudofasciolatus STL10333
Helictopleurus fasciolatus fasciolatus STL10324
Helictopleurus furcicornis STL10169
Helictopleurus cf. Furcicornis STL10332
Helictopleurus sp STL10248
Helictopleurus fasciolatus suturalis STL10234
100
80
60
40
20
0
■ A
■ S
■ R
■ A.S
■ A.R
■ S.R
■ Other